Gregory Polyakov

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Education

PhD **University of Tübingen & Max Planck Research School**, Computer Science — Tübingen, Germany, Oct 2024 – Aug 2028

- Health-NLP Lab, Supervisor: Prof. Dr. Carsten Eickhoff.
- Scholar, International Max Planck Research School for Intelligent Systems (IMPRS-IS).
- Research focus: Mechanistic interpretability of language models for NLP and IR tasks.
- Expected graduation: Aug 2028.
- Available for an internship in 2027 (flexible dates and duration).

MSc **Moscow Institute of Physics and Technology**, Applied Mathematics and Computer Science — Moscow, Russia, Sept 2021 – July 2024

- Research topics: sparse retrieval, multimodal retrieval, tool calling in LLMs.
- Thesis: "Approaches to Training Language Models for External Function Calling: Using Self-Generated Data for Better Error Detection and Recovery".

BSc **Moscow Institute of Physics and Technology**, Applied Mathematics and Computer Science — Moscow, Russia, Sept 2017 – July 2021

- Thesis: "Usage of MCMC Methods in Deep Learning".

Publications

Understanding Wacky Weights: A Dissection of SPLADE's Learned Term Importance 2026

Introduced the Wackiness Score and W-AUC to quantify semantically unrelated expansion terms across models with different vocabularies and sparsity levels. Trained 6 SPLADE variants, showing that the prevalence of wacky tokens increases with vocabulary size and that they mainly aid in-domain retrieval.

G. Polyakov, H. Scells, C. Eickhoff

arxiv.org/abs/2605.19628 (SIGIR 2026 (Reproducibility))

Interpretability Analysis of Arithmetic In-Context Learning in Large Language Models 2025

Traced how 4 LLMs (up to 12B) use in-context examples for multi-step arithmetic, showing that format consistency matters more than arithmetic correctness. Distilled in-context examples into a function vector that raises zero-shot accuracy by up to 25 points and fully recovers one-shot accuracy on Pythia-12B.

G. Polyakov, C. Hepting, C. Eickhoff, A. Bahrainian

aclanthology.org/2025.emnlp-main.92 (EMNLP 2025 (Main))

Towards Best Practices of Axiomatic Activation Patching in Information Retrieval 2025

Proposed a noise-mitigating evaluation metric for activation patching in neural rankers that filters out spurious low-confidence signals. Systematically analyzed diagnostic datasets, identifying key selection criteria for queries and documents to ensure robust interpretability analysis.

G. Polyakov, C. Chen, C. Eickhoff

dl.acm.org/doi/10.1145/3726302.3730256 (SIGIR 2025 (Short))

ToolReflection: Improving Large Language Models for Real-World API Calls with Self-Generated Data 2025
Proposed a fine-tuning strategy that enables LLMs to self-correct API calls by learning from runtime execution feedback. Co-developed evaluation datasets for tool-augmented LLMs in realistic scenarios.
G. Polyakov, I. Alimova, D. Abulkhanov, I. Sedykh, A. Bout, S. Nikolenko, I. Piontkovskaya
aclanthology.org/2025.realm-1.14 (ACL 2025 (REALM Workshop))

Sinkhorn Transformations for Single-Query Postprocessing in Text-Video Retrieval 2023
K. Yakovlev, G. Polyakov, I. Alimova, A. V. Podolskiy, A. Bout, S. I. Nikolenko, I. Piontkovskaya
arxiv.org/abs/2311.08143 (SIGIR 2023 (Short))

Experience

Huawei Noah's Ark Lab, NLP Research Engineer (full-time) Moscow, Russia
Nov 2022 – Sept 2024
1 year 11 months

- Developed a fine-tuning method that teaches LLMs to detect and recover from failed API calls (ACL 2025 REALM Workshop).
- Designed and co-implemented the end-to-end NLU pipeline for a production in-car voice assistant in Russian.
- Co-developed a Sinkhorn-based post-processing method that improves text-video retrieval quality (SIGIR 2023).

Huawei Noah's Ark Lab, NLP Research Intern (full-time) Moscow, Russia
Oct 2021 – Nov 2022
1 year 2 months

- Co-developed the reranking model for a large-scale Russian-language product search engine: improved ranking quality by 5% relative NDCG@10 and cut inference latency 1.5x by adapting ColBERTer and TinyBERT-style distillation.

Invited Talks

Understanding Wacky Weights: A Dissection of SPLADE's Learned Term Importance Glasgow IR Seminar, University of Glasgow
Oct 2026

Scholarships

Presidential Grant Scholarship Feb 2018
Feb 2018 – Aug 2021. Four-year stipend for academic and competition merit, one of 1,200 recipients nationwide.

Teaching

Teaching Assistant: Introduction to AI University of Tübingen
Winter Semester 2026/27. Conducted hands-on tutorials and evaluated student assignments.

Teaching Assistant: Modern Search Engines University of Tübingen
Summer Semesters 2025 and 2026. Conducted hands-on tutorials, evaluated assignments, and mentored student projects.

Teaching Assistant: Introduction to Python Programming University of Tübingen
Winter Semesters 2024/25 and 2025/26. Conducted hands-on tutorials and evaluated student assignments.

Skills

Programming: Python, C/C++, Bash

ML & NLP: PyTorch, Hugging Face Transformers, PEFT, LLM fine-tuning (SFT, LoRA), tool use, Scikit-learn

Interpretability: TransformerLens, activation patching, steering, probing, sparse autoencoders, cross-layer transcoders

IR: Pyserini, FAISS, sparse and dense retrieval, neural reranking, knowledge distillation

Infrastructure: SLURM, multi-GPU training (HF Accelerate), Git

Languages

Russian: Native

English: C1 (IELTS Band Score 7)